3-adic Collatz Stained Glass Visualization:

Voronoi Partitioning of the Inverse Map in Base-3 Logarithmic Spiral Space

Hiroshi Harada — April 25, 2026

Abstract

This study proposes a novel method for visualizing the geometric structure of the preimage tree of the 3-adic Collatz mapping (Collatz-3). By combining a base-3 logarithmic spiral plane with Voronoi partitioning, we reveal the deterministic topology underlying the mapping. Natural numbers n are mapped to polar coordinates (r, θ) defined by:

$$r = \log_3(n), \quad \theta = 2\pi\{\log_3(n)\}$$

The resulting Voronoi tessellation exhibits a dense-packing structure consisting of dual-helix magazines formed by $9m \pm 2$ injectors and hierarchical confluence pathways *via* $a \cdot 3^b$ launchers. Furthermore, by color-coding each cell using the cumulative distribution function (CDF) of the total stopping time (depth), the "heat, depth, and hierarchy" of the 3-adic Collatz trajectories emerge as a stained-glass-like geometric pattern.

1. Introduction

While the classical Collatz conjecture ($3n + 1$ problem) exhibits strong 2-adic characteristics, this research focuses on its 3-adic analogue, defined by:

$$f(n) = \begin{cases} n/3 & (n \equiv 0 \ (mod\ 3)) \\ (4n-1)/3 & (n \equiv 1 \ (mod\ 3)) \\ (4n+1)/3 & (n \equiv 2 \ (mod\ 3)) \end{cases}$$

Adopting the balanced ternary framework makes the mirror symmetry of ± 1 explicit through the parameter $c \in \{-1, 0, +1\}$, rendering the structural analysis of Collatz-3 geometrically intuitive.

2. Mapping to Base-3 Logarithmic Spiral Space

Natural numbers n are mapped to the polar coordinate system:

$$r = \log_3(n), \quad \theta = 2\pi\{r\}$$

This transformation visualizes the following 3-adic self-similarities:

- **Powers of 3:** aligned along a single azimuthal angle (fixed lane).
- $\boldsymbol{a \cdot 3^b}$ **launchers:** numbers sharing the same azimuthal angle.
- $\boldsymbol{9m \pm 2}$ **injectors:** clustering within specific angular bands, forming dual-helix magazines.

3. Inverse Map Structure: $9m \pm 2$ Dual-Helix Magazines

The inverse mapping is given by:

$$n_{k+1} = 3n_k - c, \quad c \in \{-1,0,+1\}$$

Numbers of the form $n = 9m \pm 2$ (non-multiples of 3) always land on a multiple of 3 in exactly one step:

$$f(9m \pm 2) \ = \ 3(4m \pm 1)$$

Moreover, the inverse mapping induces a sign inversion:

$$9m + 2 \ \rightarrow 9m - 2, \quad 9m - 2 \ \rightarrow 9m + 2$$

This toggle mechanism produces the dual-helix magazine structure.

4. Hierarchical Confluence into $a \cdot 3^b$ Launchers

For any non-multiple of 3, its multiples form a straight radial lane in the logarithmic spiral plane:

$$a, 3a, 9a, 27a, \ldots$$

The Collatz-3 inverse tree exhibits the hierarchical structure:

$$9m \pm 2 \ \text{ Magazines } \rightarrow \ a \cdot 3^b \ \text{ Launchers } \rightarrow \ 3^b \ \text{ Terminal Lane}$$

5. Dense-Packing Extraction *via* Voronoi Partitioning

Natural numbers from 1 to 3000 are plotted in the base-3 spiral space, with dummy points added to stabilize the outer Voronoi boundary. Cells for $1 \leq n \leq 1000$ are extracted and colored according to their Collatz-3 depth. To balance the visual distribution, CDF normalization is applied:

$$p(n) = \mathrm{CDF}\big(\mathrm{depth}(n)\big)$$

This equalizes the color distribution and causes the "topology of depth" to emerge as a stained-glass pattern.

6. Results

The resulting Voronoi diagram demonstrates several key features:

- **Dual-Helix Magazine Contours:** Step differences in the inverse map appear as high-contrast color shifts between adjacent cells, visualizing the toggle structure between $9m + 2$ and $9m - 2$.
- **Radial $a \cdot 3^b$ Launchers:** Lanes dominated by non-multiples of 3 appear as radial trunk-like structures.
- **Deep-Layer Nodes Filling Gaps:** Numbers with exceptionally high depth (e.g., 28, 62, and 98 for $n \leq 100$) fill the voids between primary launchers in the outer regions.

7. Discussion

The visualization confirms the following characteristics of Collatz-3:

- The $9m \pm 2$ dual-helices serve as the skeleton of the inverse tree.
- The $a \cdot 3^b$ launchers function as merger trunks for multiple trajectory series.
- Depth is densely packed in a nearly one-dimensional manner within the 3-adic space.
- A distinct hierarchical structure emerges, contrasting with the 2-adic Collatz behavior.

8. Conclusion

The inverse mapping of Collatz-3 is not merely a numerical sequence but a structure with profound geometric order. This study establishes a visualization framework combining base-3 spiral mapping, Voronoi partitioning, and CDF color-scaling. This method provides a new visual paradigm for understanding the hierarchical structure of 3-adic Collatz mappings and may be generalized to other p-adic Collatz-type systems.

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